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PALEOMAGNETISM AND POTASSIUM ARGON AGE DETERMINATIONS OF THE LASCHAMP GEOMAGNETIC POLARITY EVENT

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K/Ar ages were determined on samples belonging to the reversed formations of Laschamp and Olby (France); indicating an upper limit of 20000 years for the end of the Laschamp polarity event.

1. Introduction

Paleomagnetic and potassium argon measurements of upper Pliocene and Pleistocene volcanic rocks have led to the development of a time scale of reversals of the Earth's magnetic field (Cox, Doell, Dalrymple, Hopkins and McDougall and Chamalaun [1–6]). Polarity epochs (time interval predominantly of one polarity of the Earth's magnetic field), events (shorter time interval) and polarity transitions reflect the polarity behaviour of the Earth's magnetic field. The last epoch (Brunhes normal epoch) started 0.69 m.y. ago. It was believed to be entirely characterized by a normal polarity, as supported for the range from 0–0.69 m.y. by numerous polarity determinations and potassium argon data for volcanic rocks from Reunion, North America, Europe and Hawaii [7].

In a previous paper, the possibility of a reversal of the Earth's magnetic field in the Brunhes epoch was proposed [8] since reversed magnetizations were found in two different formations of the Chaine des Puys (Auvergne, France) volcanic chain attributed to the recent Quaternary. Also, age determinations of the lava flow of the Puy de Laschamp and of the lower flow of the Puy de Barme indicated a reversal within the Brunhes period (the Laschamp event [9]). ^{14}C dating of domitic projections overlying the Puy de Laschamp yielded ages around 8000 years [10–12]. In this paper we give the results of potassium argon determinations made on rocks belonging to these reversed formations and the related paleomagnetic data. Results for other flows of the Chaine des Puys which have abnormal directions of magnetization (likely linked to the Laschamp event) are also given.

2. Paleomagnetic data

All the lava flows studied belong to the Chaine des Puys (45°N, 3°E) volcanic chain. For sampling we applied the cap of plaster technique as well as water-cooled diamond drilling. All samples were orientated in geographical coordinates using a sun-compass. The smallest distance between two samples at a site is 3 m. Usually we selected several sites a few km apart in the same flow. After the NMR measurements we checked on the stability using Thellier's test. Then, progressive demagnetization in alternating magnetic fields was undertaken for each sample in successive steps of 50, 100, 400 Oe. The direction of magnetization remained constant while the intensity of the magnetization decreased towards zero [13]. An example of such a demagnetization curve is shown elsewhere [9]. It is a modified version of the curves

Table 1
Mean directions of magnetization

Lava flow	NMR					"Cleaned" samples				Pole position	
	N	*D*	*I*	*R*	*k*	*D*	*I*	*R*	*k*	$	
Laschamp (sites E, F)	5	221°5	−52°	4.6976	13.2	231°	−65°	4.9887	354	70°7 E	55°N
Olby (sites A, B)	6	276°5	−50°	5.2181	6.4	234°	−71°5	5.9766	214	55°E	55°N
Louchadiere	10	101°	65°5	9.1877	11	114°	57°	9.9533	193	50°7 E	11°7 S
Recoleine	7	290°	85°	4.9329	3	339°5	83°	6.9498	120	0°5 E	57°8 S
Nebouzat	6	136°	76°5	4.8748	4.4	12°	84°	5.9273	69		
Royat (19 samples in 6 sites)	6	314°	73°	5.9789	237	295°5	66°5	5.9849	331	302°5 E	47°S

N, number of samples (for Royat number of sites).
D, declination in degrees east of north.
I, inclination (positive downwards).
R, resultant of *N* unit vectors.
k, precision parameter..
$ S, N: magnetic pole in the northern hemisphere.

described by J.A.As [14] and more recently by J.D.A.Zijderveld [13]. From some samples, specimens were taken for thermal studies and intensity measurements. The results for the reversed flows are shown in table 1. The table also includes results for those flows from different volcanoes which have unexpected directions of magnetization. The position of the Virtual Geomagnetic Pole has been indicated but this does not imply the related Earth's field direction is necessarily due to a pure dipole field.

A rather large scattering in the directions can be noticed on the natural remanent magnetization. Measurements revealed that this dispersion is not only due to a viscous remanent magnetization as found by the Thellier's test but also due to a superimposed soft magnetization which was generally destroyed in a field of 250 Oe or less.

3. Potassium-argon age determination

Four samples have been selected for K/Ar-dating (see fig. 1). They belong to three different flows. At site A sample no. 443 belongs to the lower flow of the Puy de Barme (Olby flow). This flow has a reversed polarity (see table 1). At site C sample no. 456 belongs to the upper flow of the Puy de Barme. This flow has a normal polarity. (Scorias collected in a quarry of the cone are also of normal polarity.)

Sample no. 444 at site E and sample no. 445 at site F both belong to the lava flow of the Puy de Laschamp and are both reversely magnetized. They are 200 m apart. Scorias taken at the top of the volcano are also reversely magnetized and have a similar direction of magnetization [8].

To prepare unweathered fresh samples, rocks were broken with steel cone-shaped wedges. Thin sections were made to select the best specimens. The samples are basalt consisting mostly of fresh olivine, feldspar and glass.

The samples were crushed to a grain size of 0.1–1 mm. Then, the samples were cleaned with distilled water. The K-determinations were performed using flame photometry. Their analytical error is estimated to be 3%.

For the Ar-determinations, samples of about 1 g were stored in an ultrahigh vacuum line and pre-

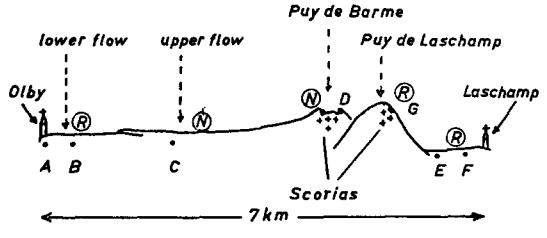


Fig. 1. Sketch drawing of sampling sites near Laschamp (see also text).

degassed at 150°C for about 10 h. The extraction system and the mass spectrometer have been described earlier [15]. The total ^{40}Ar-content was surprisingly low (approximately 10^{-8} cc/g). The main difficulty was the identification of the radiogenic component since this kind of measurement, the correction for atmospheric argon limits the accuracy. To improve this correction, we performed a large number of blank runs before and after the analysis of each sample. The amount of ^{36}Ar in these blank runs was plotted against ^{40}Ar. The interpolated curve was used to determine the required ^{40}Ar-corrections from the measured ^{36}Ar-contents in the samples. The ^{40}Ar-contents remaining was on the order of only a few percent. The error in the atmospheric ^{40}Ar-correction was, however, also a few percent. Thus, only upper limits for the K/Ar-ages can be given. The results of these runs are marked in table 2 with asterisks.

In a second series of runs the spectrometer was highly degassed. The ^{36}Ar background was so far reduced that it became possible to base the atmospheric correction on a constant ^{40}Ar/^{36}Ar-ratio of 298. For some samples the results of these measurements yielded somewhat lower upper limits for the K/Ar-ages. These results are considered more reliable.

Another method that was tried was to degass the samples in two fractions. The first step was performed at about 400°C for 10 min. About 20–30% of the total ^{40}Ar was released. Its composition was atmospheric. The partial outgassing in the first step brought about lower errors for the results of the final melting fraction. The ages for the samples 443 (Olby), 456 and 445 (Laschamp) are lower than 20000 years. In two runs of 444 (Laschamp) upper limits of 32000 years were established while two other runs of the same aliquots disclosed definitely excess ^{40}Ar. Possibly this sample contains grains which bear inherited radiogenic ^{40}Ar.

The amount of radiogenic Ar which has diffused from the sample is not serious for the present problem. If one assumes a maximum loss of 50% ^{40}Ar, the upper limit is only doubled. Even then the ages of these samples are certainly less than 50000 and 300000 years respectively. All these ages are still much lower than the K/Ar age-boundary of 0.69 m.y. for the Matuyama reversed period.

Table 2
Results of K and ^{40}Ar determination and ages from four basalts near Olby and Laschamp (Auvergne, France).

Sample	Site	Weight (g)	^{40}Ar tot. $\times 10^{-8}$ cc/g	^{40}Ar rad. $\times 10^{-8}$ cc/g	K%	Age in years
443	A	1.17	6.6	<0.13	1.52	<22000
		1.07	3.84	<0.27		<45000
		1.04	3.70	<0.15		<25000
456	C	1.07 *	8.75	<0.20	1.48	<34000
		1.55	6.70	<0.10		<17000
445	F	1.03 *	8.90	<0.11	1.67	<17000
		1.09 *	6.30	<0.10		<15000
		0.99	4.23	<0.17		<26000
444	E	1.08 *	3.60	0.95	1.57	≤153000
		1.09 *	6.58	<0.20		<32000
		1.08 *	8.0	0.70		≤110000
		1.67	4.27	<0.20		<32000

* Measured with higher ^{36}Ar background in the mass spectrometer.

4. Conclusion

K/Ar age determinations below 10^6 yr are often not possible or are less reliable since most samples have not been sufficiently degassed. They may contain inherited radiogenic Ar. In this respect, the above radiometric results are quite interesting. They show that samples exist which have inherited ^{40}Ar contents so small that K/Ar dating down to 20000 yr is feasible. The results encourage the application of the method to separated minerals of these samples and to attempt an absolute age determination instead of setting only upper limits.

The present K/Ar determinations obtained for the reversely magnetized Laschamp and Olby formations indicate an upper limit of 20000 yr for the end of the Laschamp polarity event. A lower limit of the age of the end of the Laschamp event is given by ^{14}C measurements on neighbouring volcanoes; domitic projections overlying the Puy de Laschamp and Pelat have a ^{14}C age of 8730 yr [12]. Thus, the Laschamp event appears to be very recent and could have ended between 8000 and 20000 yr. Normally magnetized lavas from Reunion [3] with K/Ar ages as young as 40000 yr indicate that the Laschamp event could not have lasted over 20000 to 30000 yr.

Further data about this reversal from other parts of the Earth would be valuable.

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